

Abdulrhman Abdulhai Salamah

AI & Machine Learning Engineer | Data Science Graduate | Certified Data Management Professional (CDMP)
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SUMMARY

First Class Honors Data Science graduate ranked 1st in cohort at King Saud University and Certified Data Management Professional (CDMP). AI Engineer with 2+ years of freelance experience building fully autonomous agentic AI systems, Retrieval-Augmented Generation (RAG) pipelines with LangChain, and customized machine learning models for companies and website owners — alongside high-impact roles at Saudi Aramco (AI-driven predictive maintenance) and the Digital Government Authority. Recipient of the 2025 'Ideal Student' award, combining hands-on LLM and agent engineering with strong Python, R, and SQL foundations.

EXPERIENCE

Freelance AI Engineer & Machine Learning Developer, Self-Employed <i>Remote</i>	06/2023 – 01/2026 Riyadh, Saudi Arabia
- Delivered AI solutions to multiple companies and website owners over 2 years and 7 months while completing a full-time degree, ranked 1st in cohort. - Designed and deployed fully autonomous agentic AI systems with tool/function calling to automate customer support, business workflows, and data processing. - Engineered Retrieval-Augmented Generation (RAG) systems using LangChain and vector databases, grounding LLM responses in client knowledge bases to eliminate hallucination. - Trained and fine-tuned customized machine learning models for client-specific prediction, classification, and recommendation use cases. - Integrated LLM APIs (OpenAI GPT-4o) with clients' internal APIs, websites, and e-commerce platforms to build end-to-end automation systems.	
Data Scientist, Saudi Aramco <i>Power Systems — Predictive Maintenance</i>	09/2025 – 12/2025 Dhahran, Saudi Arabia
- Analyzed and visualized operational data from industrial plants and integrated data platforms to support reliability engineering. - Developed machine learning models to forecast equipment alarms and failures, enabling AI-driven predictive maintenance; earned 2 certificates of appreciation.	
Data Science Lead / Machine Learning Engineer, Emerging Technologies Club at KSU	09/2024 – 12/2025 Riyadh, Saudi Arabia
Led a team building multiple machine learning models for competitions and hackathons, coordinating model development end to end.	
Data Scientist / Machine Learning Engineer, Digital Government Authority	06/2025 – 08/2025 Riyadh, Saudi Arabia
Developed machine learning components and contributed to backend development of an AI-powered accessibility tool that makes government documents accessible.	
Member, Emerging Technologies Club at KSU	05/2024 – 09/2024 Riyadh, Saudi Arabia
Active member participating in hackathons, conferences, and club events.	

EDUCATION

Bachelor of Science in Applied Information Systems — Data Science Track, King Saud University 08/2021 – 01/2026
- GPA: 4.80/5.00 — Excellent with First Class Honors; ranked 1st out of 21 graduates in the major. 'Ideal Student' Award (2025) for the entire College of Applied Computer Sciences. - Dean's Award for Excellence — three consecutive years (1444–1447 AH). 1st place — Applied Computer Science Graduation Projects competition (Emma'a).

SKILLS

Technical Skills: Generative & Agentic AI: LLM APIs (OpenAI GPT-4o), LangChain, RAG, Vector Databases & Embeddings,	Soft Skills & Professional Competencies: - Team Leadership - Project Management
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- Autonomous AI Agents, Tool/Function Calling, Prompt Engineering, Multi-Agent Systems
- **Machine Learning:** Supervised & Unsupervised Learning, Deep Learning, NLP, Computer Vision, Feature Engineering, Model Evaluation, Model Fine-Tuning
- **Programming Languages:** Python, Java, R, C++, SQL, PHP
- **Tools & Libraries:** PyTorch, Scikit-Learn, NumPy, Pandas, Streamlit
- **Data Visualization:** Power BI, Matplotlib, R
- **Databases:** MySQL, Vector Databases
- **Web Development:** HTML, CSS, JavaScript, REST APIs
- **Version Control:** Git, GitHub
- **Certification:** CDMP (Certified Data Management Professional)
- Problem-Solving
- Critical Thinking
- Decision-Making
- Communication
- Time Management
- Adaptability
- Teamwork
- Fast Learner

PROJECTS

AI-Powered Customer Support Ecosystem — Freelance (E-commerce Gaming Store) 2025

- Took full ownership of designing and implementing an intelligent customer support automation architecture for a high-volume e-commerce store specializing in games and gaming hardware, handling complex queries such as game compatibility, digital key delivery troubleshooting, and hardware recommendations.
- Architected the core engine on OpenAI API models as the backbone of the chatbot interface for intent recognition and conversational flow.
- Engineered a full RAG pipeline with LangChain: vectorized the entire product catalog, FAQs, and troubleshooting guides into a vector database for accurate, context-grounded responses.
- Designed autonomous GPT-4o tool-calling agents that decide when to query the internal order/digital-key tracking API, parse the payload, and return real-time status updates to customers.
- Deflected a significant share of tier-1 support tickets and delivered 24/7 instant resolution for digital product delivery issues while strictly anchoring the LLM to a verified database.

Emma'a — إيما (Graduation Project — 1st Place)

- Trained two computer vision models to recognize and classify letters and words of Arabic Sign Language, then integrated both models into a single software product.

Murshed 04/2025 – present

- AI-based assistant designed and developed with colleagues during the Transformation Hackathon at King Saud University.

Kolna Akther 01/2025 – 02/2025

- Program that detects potholes and objects on roads using machine learning.

Advertisement Program 01/2025 – present

- Developed a program to automate advertising on social media.

AI Assistant — Jarvis 05/2024 – present

- Personal AI assistant with system-level access to a PC, automating daily tasks.

Product Success Prediction 01/2024 – 03/2024

- Machine learning model in R predicting new product success from historical sales data.

CERTIFICATES

Agentic AI Engineering Workshop (In Progress)
SDAIA

SDAIA Bootcamp: Building AI Applications Using Python
SDAIA & KSU

Machine Learning Specialization
DeepLearning.AI

Generative AI for Everyone
DeepLearning.AI

Intro to AI Engineering
Scrimba

Python Data Structures
University of Michigan

Using Python to Access Web Data
University of Michigan

Data Analysis with R Programming
Google

Foundations: Data, Data, Everywhere
Google

Introduction to Responsible AI
Google

Responsible AI: Applying AI Principles with Google Cloud
Google

Business Implications of AI: Full Course
EIT Digital

AWARDS

- 'Ideal Student' for the entire College of Applied Computer Science (2025) — KSU.
- Dean's Award for outstanding/distinguished students — 1444–1445, 1445–1446, 1446–1447 AH — KSU.
- 1st place — Applied Computer Science Graduation Projects competition (Emma'a) — KSU.
- 2 Certificates of Appreciation — Saudi Aramco.

References available on request.